

BDG: Batch-Dispersion Guidance for Property-Band Generation with Continuous Flow Matching

A feedback-servoed deviation gain for property-gradient guidance, tested on QM9 3D molecules and on DNA enhancers on the probability simplex

Frame the study

Problem. Generate 3D molecules whose property lands inside a tolerance band around a target, without retraining the generator. A design campaign wants many usable candidates, so the quantity that matters is **in-band**: the fraction of samples inside the band.

Central question. Guidance concentrates a batch toward the target, and that dispersion is a by-product of its strength. Does making dispersion a quantity the user *requests* raise band coverage?

Hypothesis. At a centred target, coverage is limited by batch spread rather than centring error, so a calibrated spread setpoint should raise it.

MODALITY 1

QM9 3D molecular coordinates [1]

State space $\mathbb{R}^{N \times 3}$ with a relaxed atom-type simplex, $N \leq 29$. Invariant to translation, rotation and atom permutation [2].

MODALITY 2

DNA enhancers on the probability simplex [3]

State space $(\Delta^3)^L$, one simplex per position over {A, C, G, T}. No rotation group; the constraint is that each position stays on the simplex.

Takeaway. These are different state spaces, not two datasets of the same kind: one is a Euclidean coordinate space with a geometric group acting on it, the other a product of probability simplices.

Datasets, splits, and leakage

MODALITY 1

QM9 [1]: 133,885 molecules, explicit hydrogens, pinned by SHA-256

Split	Used for	n
train_a	generator and guide f_A	51,527
train_b	evaluator f_B only	51,527
val	checkpoints, atom counts	17,748
test	held out	13,083

How leakage is prevented. f_A and f_B are trained on **disjoint halves**, so the network that scores us never saw the data that trained the network that steers us. Normalisation constants come from train_a alone. The calibration set that fixes δ is disjoint from the sets supplying targets and atom counts.

MODALITY 2

Kenyon-cell enhancers [3]: 6,126 *Drosophila* sequences

The evaluator is an exact count, so no train/validation split is needed and no leakage is possible.

We do not merge the two species distributed together. Their GC distributions differ, so steering spread on the mixture would partly steer the species-mixing proportion, which is the quantity under study.

Preprocessing (M1)

Coordinates zero-centred per molecule, in Å; dense [B, 29, ·] padding with a validity mask; DFT properties in D / Bohr³ / Ha.

Representations, conditioning, handling

Representations, and why

M1: coordinates plus a *relaxed* one-hot over {H, C, N, O, F}. The continuous relaxation is what lets one continuous flow carry both geometry and discrete types.

M2: a point of $(\Delta^3)^L$ per sequence. The simplex counts as a continuous state space by the assignment's definition, and it is the native geometry of the sequence literature [4, 5].

Conditioning

Base generation is **unconditional**. The property enters only at sampling time, through guidance.

This is deliberate: a conditional model would give the controller nothing to steer, and would break the M1/M2 parallel we are testing.

Modality-specific handling

M1 invariance: translation by projection onto the zero-centre-of-mass subspace; rotation by messages depending only on squared pairwise distances, with coordinate updates along difference vectors; permutation by symmetric pooling. Verified numerically: **12/12 gates pass to $\sim 1e-17$.**

M2 simplex mapping and decoding: sequences are relaxed onto the simplex and decoded by per-position argmax; the property is scored by an **exact** GC count on the decoded sequence.

Flow-matching baseline: the formulation

Data x_1 , source x_0 ; time t runs from 0 (noise) to 1 (data).

Probability path

linear interpolant [6, 7, 8]

$$x_t = t x_1 + (1 - t) x_0 \qquad u_t(x_t \mid x_0, x_1) = x_1 - x_0$$

Source is not isotropic

Coordinates $x_0^c = \Pi_0 \varepsilon$, with Π_0 the projection onto the zero-centre-of-mass subspace of the valid atoms, of dimension $3(N - 1)$. Type channels are a masked Gaussian.

Loss, exactly as implemented

masked mean over valid entries,
two heads normalised separately,
 $\lambda = 1$

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E} \left[\frac{1}{3 \sum M} \|M \odot (v_\theta^c - u^c)\|^2 + \lambda \frac{1}{K \sum M} \|M \odot (v_\theta^f - u^f)\|^2 \right]$$

New symbols: M the atom mask, K the number of type channels, v_θ the learned velocity. Separate normalisation stops padding diluting the loss.

Endpoint estimate

closed form at every t

$$\hat{x}_1 = x_t + (1 - t) v_\theta(x_t, t)$$

The object that every guidance arm differentiates.

Flow matching: training, selection, sampling

Architecture [9]

Dense E(n)-equivariant graph network: width 256, 8 layers, SiLU, **3,753,229 parameters**. Time enters at the input embedding. No attention.

Optimiser, schedule, budget

Adam, lr 2e-4, cosine annealing to 5% of it, batch 256, gradient-norm clip 5.0, EMA decay 0.9999 with warm-up; sampling always uses the EMA copy. **1500 epochs = 303k steps**, 11.4 h on one B200 MIG slice, seed 20260918.

Model selection (pre-registered)

Highest *validation* atom stability at NFE 100, ties broken on validation loss, evaluated every 25 epochs.

Disclosed amendment. The rule selected epoch 1300; we ship **epoch 1500**, because the 512-sample selection signal could not resolve the last 200 epochs. The 3-seed × 10,000 benchmark confirms it on every metric: +0.0018 atom, +0.0097 molecule, +0.0110 validity, same sign on all three seeds.

Sampling [6]

Probability-flow ODE, **explicit Euler**, uniform grid, **100 steps ⇒ NFE 100**.

Quality saturates by 500 steps (100 → 500 buys 0.65 atom and 2.8 molecule points), so 100 is a cheap operating point.

Takeaway. The closed-form endpoint estimate at every t is why this family is convenient for guidance, and the reason we carry it forward in 2a.

Diffusion baseline: the formulation

Continuous-time variance-preserving process [2, 10, 11]; τ runs from 0 (clean) to 1 (noise), linear β .

Noise schedule

$$\beta(\tau) = \beta_{\min} + \tau (\beta_{\max} - \beta_{\min}), \quad \beta_{\min} = 0.1, \quad \beta_{\max} = 20$$

Forward process

$$x_{\tau} = \alpha(\tau) x_0 + \sigma(\tau) \varepsilon, \quad \log \alpha(\tau) = -\frac{1}{2} \left(\beta_{\min} \tau + \frac{1}{2} (\beta_{\max} - \beta_{\min}) \tau^2 \right), \quad \alpha^2 + \sigma^2 = 1$$

Prediction target: ε

Same two-term masked-mean loss as the flow model, with coordinate noise projected onto the zero-centre-of-mass subspace so the target is equivariant. Times $\tau \sim \text{U}(1\text{e-}3, 1)$ keep σ away from zero.

Reverse sampler

probability-flow ODE

$$\frac{dx}{d\tau} = -\frac{1}{2} \beta(\tau) (x + s_{\theta}(x, \tau)), \quad s_{\theta} = -\varepsilon_{\theta} / \sigma$$

Integrated backwards on a decreasing 100-step grid, then a jump to the Tweedie posterior mean.

Derived quantities guidance consumes

$$\hat{x}_0 = (x_{\tau} - \sigma \varepsilon_{\theta}) / \alpha, \quad k = \sigma^2 / \alpha$$

Verified: $\alpha^2 + \sigma^2 = 1$ to 6 decimal places, $\alpha(0) = 1$, $\alpha(1) = 0.0066$.

Diffusion: training, selection, sampling

Matched by construction. Same representation, split, **EGNN backbone**, optimiser, learning-rate schedule, batch size, EMA, gradient clipping, seed and **pre-registered selection rule** as the flow model. It differs **only in the corruption process**, which makes 2a a comparison of families rather than of engineering.

Architecture and training [9]

EGNN 256×8, 3.75M parameters, ϵ -prediction head. Adam 2e-4, cosine to 5%, batch 256, clip 5.0, EMA 0.9999, 1500 epochs.

Model selection

Highest validation atom stability at NFE 100, ties on loss: the identical rule.

Sampling [2, 11]

100-step decreasing grid, then the Tweedie endpoint jump, so the unguided budget is **NFE 101** against the flow model's 100. We report that asymmetry rather than hide it.

Status, stated plainly. Our own VP diffusion is implemented, gated and smoke-tested; the full-scale run is **[TODO: v3 Job A]**. The only *measured* diffusion comparator today is a borrowed released checkpoint, and 2a labels it as borrowed. We will not claim a family winner from it: 2a states what the evidence does and does not support.

Metrics, and what each cannot tell us

Control metric: in-band

Fraction of samples with $|f_B(x) - y| \leq \delta$, scored by the **held-out** evaluator. $\delta = 2 \times f_B$'s calibration MAE on 3,000 molecules disjoint from the target and atom-count sets: **pre-registered**, so it cannot be chosen after seeing results. Reported both continuous and decoded.

Decomposition

$RMSE^2 = \text{bias}^2 + \text{spread}^2$, both in units of δ , plus spread relative to the unguided control. Spread is the quantity the innovation targets.

Quality and diversity

Atom stability, molecule stability, validity, uniqueness, novelty, pairwise diversity, under EDM conventions [2], with bond tables verified **entry-for-entry** against the reference implementation.

Sample counts

Base models: 3 seeds $\times$ 10,000. Guidance cells: n per cell $\times$ 3 seeds, $n = \text{[TODO: v3 n]}$. Screening cells: $n = 256\text{--}512$, labelled wherever used.

Limitation we state up front. in-band rewards *any* arm that contracts the batch, including onto chemically implausible structures, so every in-band number in this deck appears beside the chemistry it spent. No arm is ranked on in-band alone: a method that wins coverage by destroying molecules has not won.

Fairness: what is matched, what is not

Factor	Matched	Disclosed mismatch
Data, split, preprocessing, conditioning	identical; unconditional for both	none
Atom counts, evaluator, metric code, δ	identical, EDM conventions [2]; atoms from train_a	none
Starting noise, seeds	shared, so arms are paired	nothing paired across backends
Backbone and parameters	same EGNN 256×8; 3.75M vs 3.75M	borrowed comparator: 256×9 + attention, 5.34M
Sampling budget	100 steps for both	diffusion needs NFE 101 for its endpoint jump
Training compute	identical recipe	[TODO: v3 Job A]; borrowed checkpoint budget unknown
Feature scaling	ours: unscaled one-hot	borrowed comparator scales by 1/4

Guidance-strength fairness

The headline fixes **w = 1** for every arm: what each rule does at one dial setting, not which is best at its own optimum. **Equal w is not equal force**: the applied-correction share spans 5.8× across the arms where it can be measured, so every headline number carries "at w = 1".

Statistics

Binomial standard errors; paired comparisons within a backend; multiplicity-corrected significance beside the single-test value, since each arm is selected over a grid. in-band is **never compared across backends**: δ is set by that backend's evaluator.

Choose the base model

Model	Source	Atom stab. ↑	Mol. stab. ↑	Validity ↑	Uniq. ↑	NFE ↓	Params
Flow matching (ours)	reproduced	0.9366 ±.0012	0.3993 ±.0085	0.7625 ±.0031	0.9939	100	3.75M
VP diffusion (ours)	reproduced	[TODO: v3 Job A]	[TODO]	[TODO]	[TODO]	101	3.75M
EDMsecond [2]	borrowed ckpt.	0.9644	0.6725	0.8490	[TODO]	100	5.34M
EDM, published [2]	quoted	0.9870	0.8200	0.9190	—	1000	—
EDM, unscaled one-hot [2]	quoted	0.9570	0.4690	—	—	1000	—
Real QM9 (ceiling)	measured	0.9940	0.9560	0.9820	1.0000	—	—

§4.2 table. QM9, unconditional, 100-step Euler probability-flow ODE, our evaluator under EDM conventions. Our flow row is 3 seeds × 10,000 samples; EDMsecond is a single-seed gate run at n = 2,000, so it is not directly comparable.

Carried forward: flow matching, and not for a fidelity win

The matched own-diffusion row is [TODO: v3 Job A], and the only measured diffusion comparator is a borrowed checkpoint differing in architecture, training half and feature scaling, so it cannot decide the question. Flow matching's linear path gives a **closed-form endpoint estimate at every t**, the object every guidance rule differentiates, at NFE 100 against a 1000-step published protocol.

Our stability gap, in rank order

- 1. Unscaled one-hot** leads: the published ablation removing exactly that scaling costs 3 atom and 35 molecule points [2].
- 2. Training budget:** 303k steps vs ~1.56M. **Ruled out:** the half-data split (0.33 atom points) and the evaluator (bond tables match entry-for-entry).

Guidance strategy

Rule and target

$$G_i = \frac{\text{num}_i}{s^2} J_i^\top g_i, \quad \text{num}_i^{\text{plug}} = y - F_i, \quad F_i = f_A(m_i), \quad g_i = \nabla_m f_A(m_i)$$

A frozen property predictor f_A is differentiated at the endpoint estimate m_i , and the gradient is pulled back to the sampling variable. Target y is the **q50** value of the property, one fixed number per property. A DPS-style plug-in Gaussian-energy field [12].

Trained, pretrained, frozen

Generator $v\theta$: trained by us, frozen at sampling.

Guide f_A : trained by us on train_a; frozen, eval(), gradients disabled.

Evaluator f_B : trained on train_b; frozen, and it never participates in sampling.

Nothing is trained at guidance time.

Strength schedule, normalisation, clipping

Strength: correction converted by $w \cdot (1 - t)/t$, w the guidance weight [12, 14].

Normalisation: denominator s^2 , with s the guide's property standard deviation.

Trust region: G_i rescaled so $\|G_i\| \leq \|v_i\|$ per sample, after w . Prior practice [13], shared by every arm.

Window: guidance on for $t \geq 0.5$ only.

Takeaway. The trust region is load-bearing, not cosmetic: removing it produces **222 non-finite samples** where the clipped run produces **0**.

Guidance works, and what it costs

Arm	in-band ↑	Spread / unguided	Mol. stab. ↑
unguided	.090 / .121 / .051	1.00 / 1.00 / 1.00	.434 / .434 / .434
plug-in (best w) [12]	.184 / .219 / .133	0.52 / 0.60 / 0.47	.246 / .199 / .320
TFG (w = 4, μ only) [15]	.535 / – / –	0.22 / – / –	.172 / – / –
BDG (w = 32, τ_{mult} = 0.5)	.180 / .227 / .106	0.49 / 0.68 / 0.53	.348 / .262 / .340

§4.3 table. Each cell is μ / gap / α , at q50, otherwise matched: one generator, one seed, n = 256, best strength per arm over $w \in \{0.05, \dots, 32\}$, no chemistry floor. Three-seed confirmation: **[TODO: v3]**

The target improves

Plug-in: **2.0×** on μ , **1.9×** on gap, **2.6×** on α over unguided. TFG reaches **6.0×** on μ .

The cost is real

Molecule stability falls from 0.434 unguided to 0.246 (plug-in, μ) and to 0.172 (TFG). Validity falls with it.

The mechanism is visible

Spread contracts from 1.00 to 0.22–0.60. Guidance works by concentrating the batch; the arm that contracts most wins in-band and loses the most chemistry.

Takeaway. That trade-off is inherited, not chosen. 2c asks whether the user can choose it instead.

Motivating the innovation

1–2

band widths from the target
(centring error)

5–9

band widths of batch spread

The weakness. At a centred target, coverage is a *dispersion* problem: guided batches sit 1–2 band widths from the target while their spread is 5–9 band widths. Yet every guidance rule optimises a *centring* objective and exposes **one dial, w** , which moves centring and spread together in a ratio the user never chooses.

The hypothesis. If the sampler *measures* its own batch dispersion and servoed toward a *requested* value, the user trades spread against centring deliberately, in the units of the property rather than in units of push.

Prior art	What it already owns
CFG-Ctrl [16], Feedback Guidance [17]	the guidance scale read as a control gain
MGD [18]	ensemble moments driven to a target, two-sided
Variance-Tilted Diffusion [19]	batch-spread targeting; widen-only, fixed linear feature
Particle Guidance [20]	batch-coupled repulsion, no setpoint
Negative-guidance window [21]	negative weight widens the distribution

We claim none of these ingredients

In a bounded search (about eight queries, abstracts only, 25 Sep 2026) we did not find a scalar guidance weight set by the batch variance of a *learned property predictor's own endpoint prediction*, against a *two-sided, user-chosen setpoint, signed so the weight passes through zero*. That bounds our claim; it is not a priority claim.

BDG: the mechanism

Two batch statistics and a scale-free error (per guided step, batch B)

$$\bar{F} = \frac{1}{B} \sum_i F_i \quad V_b = \frac{1}{B-1} \sum_i (F_i - \bar{F})^2 \quad e = \frac{V_b - \tau^2}{\tau^2}$$

Signs are the whole mechanism

$V_b > \tau^2 \Rightarrow$ **contract**

$V_b < \tau^2 \Rightarrow$ **expand**

$V_b = \tau^2 \Rightarrow$ the term vanishes, and BDG is exactly the plug-in rule [12]

The rule

$$\begin{aligned} \text{num}_i &= (y - F_i) - \eta e (F_i - \bar{F}) \\ &= \underbrace{(y - \bar{F})}_{\text{centring, gain } w} - \underbrace{w_{\text{eff}} (F_i - \bar{F})}_{\text{deviation, gain } w_{\text{eff}}}, \quad w_{\text{eff}} = 1 + \eta e \end{aligned}$$

$$\frac{\partial V_b}{\partial F_i} = \frac{2}{B-1} (F_i - \bar{F})$$

A descent direction on the thing that binds. The chain rule carries it to m_i along g_i ; the $2/(B-1)$ is absorbed into η , so the gain does not scale with B.

Cost: zero extra NFE

Both terms multiply the same g_i , so BDG shares the one backward pass the plug-in rule already performs. Cost accounting is byte-identical.

Trainable: nothing

Generator and f_A stay frozen. **Two new hyperparameters:** the gain η and the setpoint $\tau = \tau_{\text{mult}} \cdot s$, a fraction of the guide's own property standard deviation.

BDG: the reduction, and its bounds

We state the reduction ourselves: num_i is affine in F_i , so it factors exactly

$$(y - F_i) - \eta e (F_i - \bar{F}) = (1 + \eta e) (y_{\text{eff}} - F_i), \quad y_{\text{eff}} = \frac{y + \eta e \bar{F}}{1 + \eta e}$$

Verified to **3.0e-15** analytically and **6.5e-7** on the real field.

What that means

At any instant BDG is the **plug-in field at a rescaled weight aiming at a shifted target**. It adds no new direction: any guidance whose per-sample coefficient is affine in F_i is the plug-in rule reweighted.

What it does not remove

w_{eff} is set **online** from a measured quantity and may **change sign**, which w cannot do without also reversing the centring term.

Bound, for any batch

$$V_b \geq 0 \Rightarrow e \geq -1 \Rightarrow w_{\text{eff}} \geq 1 - \eta$$

This licenses running two-sided with no clamp. **Design constraint $\eta > 1$** , or w_{eff} never goes negative and the controller cannot expand at all.

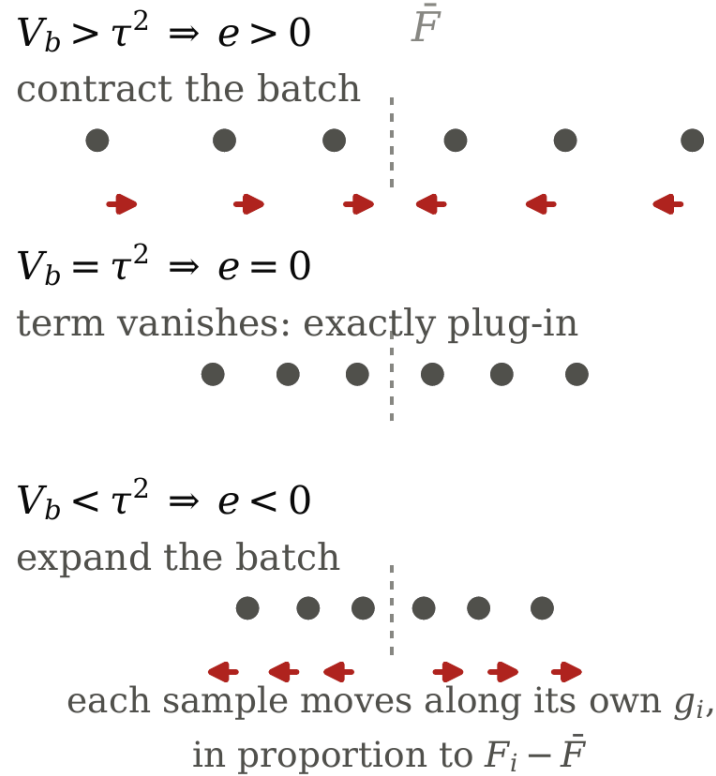
Equilibrium, stated honestly

$$w_{\text{eff}} = 0 \Rightarrow V^* = \tau^2 (1 - 1/\eta)$$

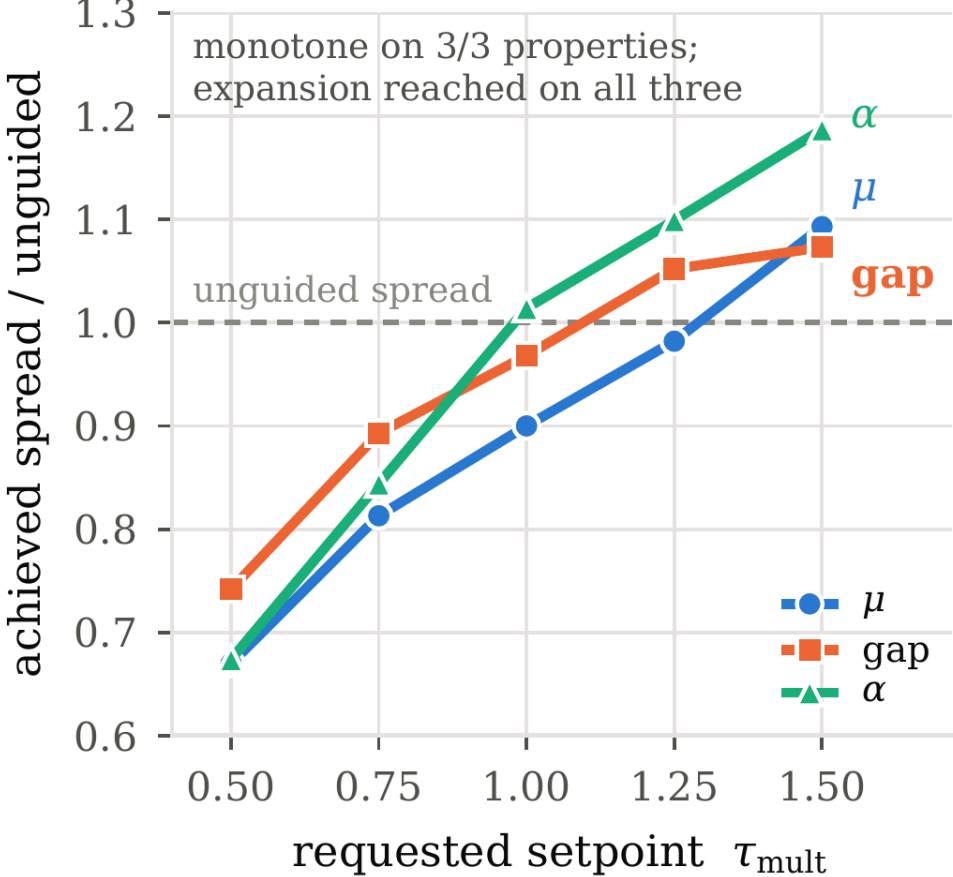
An **asymptote the 50-step window never reaches**. We never say the controller holds its setpoint, and we order every table by *measured* w_{eff} .

The controller, measured

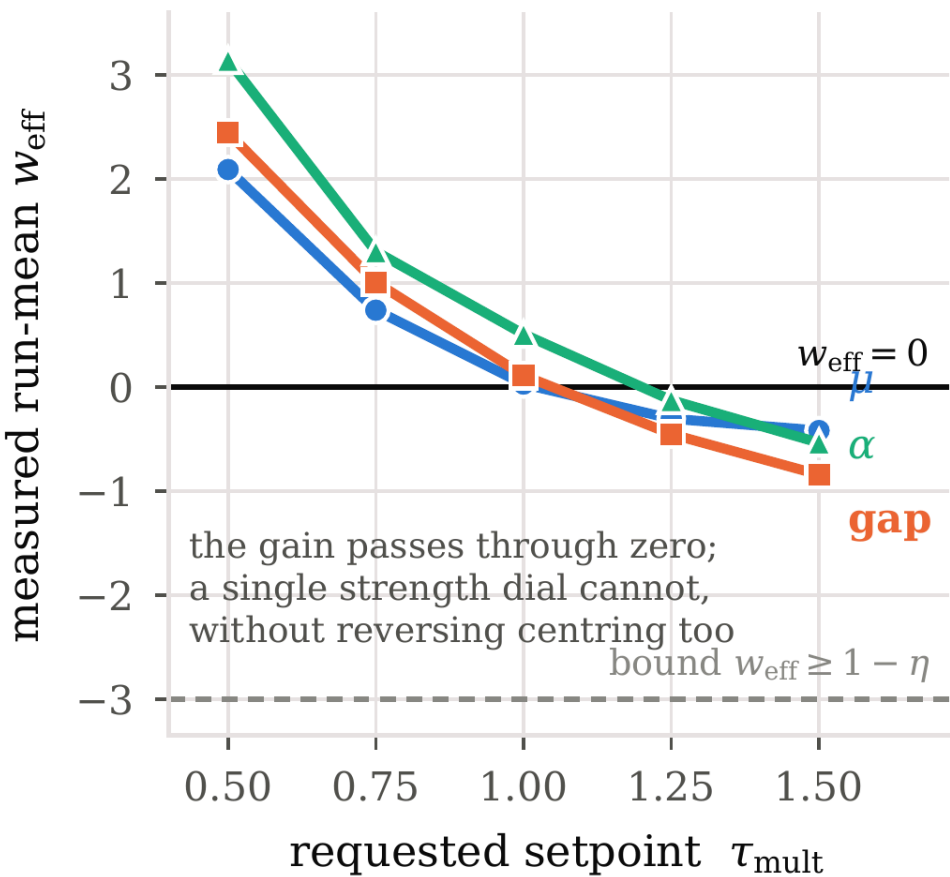
(a) the sign mechanism



(b) setpoint controls achieved spread



(c) the deviation gain changes sign



(a) The sign of e is the whole mechanism, and $e = 0$ recovers the plug-in rule exactly.

(b) The requested setpoint maps **monotonically** onto achieved spread on **3/3** properties, and crosses *above* the unguided spread on all three, which the strength dial does not reach.

(c) The measured deviation gain **passes through zero**, inside its bound $w_{eff} \geq 1 - \eta$. A single strength dial cannot do that without reversing the centring term as well.

Testing the innovation: ablations

§4.4 table. BDG ablations at q50, ordered by *measured* w_eff, never by τ_{mult} . Port cells: n = 256, one seed, w = 1, own unguided control.

Variant	Role	η	τ_{mult}	Spread / ung.	in-band ↑	Mol. stab. ↑
plug-in, w = 1	base	–	–	0.714	0.1094	0.3633
BDG $\eta = 0$	gate	0	1.0	0.714	0.1094	0.3633
BDG one-sided, asked to expand	gate	4	1.5	0.714	0.1094	0.3633
BDG contract	ladder	4	0.5	0.670	0.1406	0.3359
BDG neutral	ladder	4	1.0	0.900	0.0820	0.4258
BDG expand	ladder	4	1.5	1.093	0.0703	0.4141
plug-in, w = 4	control	–	–	0.584	0.1406	0.2656

Gates prove the implementation; they are not controls

$\eta = 0$ is **bit-identical** to the plug-in rule, so the dispersion term is the only addition.

The one-sided variant asked to expand is also bit-identical (raw e = −0.984 clamped to 0), so the sign branch is real and the clamp is self-consistently inert.

Control: open-loop e replay
Fresh noise, $\eta = 4$, $\tau_{\text{mult}} = 0.5$: recovers **69.5–96%** of the closed loop.

Control: signed-w plug-in
[TODO: not run; drop rule applies]

All variants: three-seed v3 grid
[TODO: $\eta \in \{0,1,2,4,8\} \times \tau_{\text{mult}} \in \{0.5,0.75,1,1.5\}$, w $\in \{1,4\}$]

What the ablations establish

Comparison	What it rules out
$\eta = 0$ vs plug-in	that anything other than the dispersion term changed
one-sided vs two-sided	that the expansion branch is decorative
the τ ladder	that the setpoint does not reach achieved spread
plug-in at $w = 4$	that the coverage came from controlling rather than pushing
open-loop e replay	that the effect is feedback rather than a time-varying schedule
signed-w plug-in [TODO]	that the same spread is not available more cheaply

1 · The setpoint controls spread

Monotone on **6 of 6** curves: 0.670 $\rightarrow$ 1.219 on the guide's scale, 0.650 $\rightarrow$ 1.197 on the evaluator's, reaching expansion above 1.0 on all six, which the strength dial does not reach. w_{eff} changes sign up to **34 times** in the 50-step window and spans $[-2.64, +10.35]$ against the bound $w_{\text{eff}} \geq -3$.

2 · Coverage does not improve

No rung beats plug-in at $|z| \geq 3$: over **72 paired tests** the largest is 3.07, which a Šidák correction turns into **p = 0.14**. Removing the chemistry floor does not change this.

3 · The trade-off moves, not the coverage

At *matched* in-band of 0.1406, BDG keeps molecule stability **0.3359** against plug-in's **0.2656**.

Honest reading. The schedule largely transfers across seeds (69.5–96% recovered open-loop), and feedback adds a small measurable remainder (0.3–6.6%). We do not claim the loop is irreducible to a schedule.

Position against recent work

Method	Status	in-band ↑	Mol. stab. ↑	Diversity ↑	Gen. passes	Guide passes
DPS-style plug-in [12]	reproduced	[TODO: v3]	[TODO]	[TODO]	10,000	4,000
TMPD-inspired [22]	reproduced	[TODO: v3]	[TODO]	[TODO]	10,000	4,000
LGD-MC [23]	reproduced	[TODO: v3]	[TODO]	[TODO]	12,000	20,000
TFG [15]	reproduced	[TODO: v3]	[TODO]	[TODO]	8,000	20,000
D-Flow [24]	reproduced	[TODO: v3]	[TODO]	[TODO]	[TODO]	[TODO]
Flow matching, unguided	internal ref.	[TODO: v3]	[TODO]	[TODO]	10,000	0
BDG (ours)	reproduced	[TODO: v3]	[TODO]	[TODO]	10,000	4,000

§4.5 table. All reproduced inside our pipeline on our frozen checkpoint, guide and evaluator, at q50 and w = 1. Passes are per cell at n = 5,000.

Why these five

All are inference-time guidance published in the last five years, all steer a frozen generator with a frozen predictor, and all run on *our* checkpoint. That isolates the **guidance rule** from the generator, which is what our contribution is about.

Efficiency, measured not inferred

BDG's two terms share one backward pass: **4,000** guide passes per cell against **20,000** for LGD-MC and TFG, a fifth of their guidance cost at the same generator budget. BDG is *not* the cheapest overall: TFG uses fewer generator passes.

Fair study, and fair discussion

Reproduced versus quoted

Every row in the 2e table is **reproduced by us** on one frozen checkpoint, one guide, one evaluator, one target, one sample count. The only *quoted* numbers anywhere in this study are the published base-model rows in 2a, labelled as quoted and never pooled with ours.

Three port caveats we disclose

Plug-in is a DPS-style Gaussian-energy field; the original DPS differentiates the residual L2 norm [12].

TMPD is the local-linear scalar limit of the published method: a related-work comparison, not a faithful nonlinear port [22].

TFG runs at its published QM9 hyperparameters but through our deterministic ODE sampler (the original used a stochastic DDIM update), and our unscaled one-hot features move about 16× less relative to coordinates than the released checkpoint's. **Never read these numbers against TFG's published table** [15].

Better

Chemistry retained at matched coverage. Guidance cost: a fifth of LGD-MC's and TFG's guide passes.

Comparable

in-band against the plug-in rule: the honest reading of the ablation null.

Worse

TFG reaches far higher in-band (0.535 vs 0.180 on μ) at a chemistry cost we judge unacceptable (0.172 vs 0.348), and it uses fewer generator passes.

[TODO: v3 completes this comparison]

The transferred method

What stays the same

The **whole controller**: $\bar{F}$, V_b , the error e , the setpoint τ , the gain η , the bound $w_{\text{eff}} \geq 1 - \eta$, and the reduction. **Transferred with zero code change.**

Why it can: BDG's step is a per-sample scalar times the plug-in step, so it inherits that step's tangency and needs **no covariance matrix and no inverse**. That matters here, because the natural simplex covariance $\text{diag}(p) - pp^T$ is **singular**.

What must change	Modality 1 (QM9)	Modality 2 (DNA) [4, 5]
State space	$\mathbb{R}^{N \times 3}$	$(\Delta^3)^L$
Generator	EGNN	linear-path flow model on the simplex
Property	trained equivariant network	analytic, differentiable relaxation of GC content
Evaluator	f_B	exact GC count
Decoder	none	per-position argmax

Takeaway. The transfer tests the *controller*, not the property network: that is the point of changing the property from a learned net to an exact function.

Training procedure (M2)

Linear-path flow matching on $(\Delta^3)^L$, trained on all 6,126 Kenyon-cell sequences [3], validation-based early stopping, checkpoint selected on validation loss. Architecture and budget: **[TODO: v3 Job B]**.

Sampling procedure (M2)

Same solver family and step count as M1, guidance on the same window, same velocity-relative trust region, decode by per-position argmax. The clip fired on **0–3%** of steps; the expanding cell clipped **once in 25,600**.

Transfer results

Structural signature	QM9	DNA simplex
$\eta = 0$ reproduces the plug-in rule	bit-identical	bit-identical
one-sided at an expanding setpoint	exact	exact, $w_{\text{eff}} = +1.000$
setpoint ladder monotone in spread	$0.67 \rightarrow 1.22$	$0.81 \rightarrow 1.01$
expansion reached	1.168	1.006
cost identical to the plug-in rule	yes	yes, byte-identical
contraction breaks the fidelity floor	yes	yes
in-band improves	no	no

vs recent simplex models	Dirichlet FM [4]	Fisher Flow [5]	MOG-DFM [3]	BDG (ours)
GC in-band, DNA simplex	[TODO: v3 Job B]	[TODO: v3 Job B]	[TODO: v3 Job B]	[TODO: v3 Job B]

Fair discussion. Every structural signature transfers; band coverage improves on neither modality; the external comparison is **[TODO: v3 Job B]**.

Not the published benchmark

Standard protocol: class-conditional generation at **500 bp** on 81-/47-class sets of ~90,000 sequences, scored by FBD over 10,000 samples. Ours: GC-band coverage, **6,126** sequences, **200 bp**. Our base generator is the linear-FM baseline those papers report, but the numbers are not comparable and are never pooled.

Further disclosures

Gumbel-Softmax FM [25] does not run the enhancer benchmark, so only two of the three closest named methods apply.

Fisher Flow [5] disputes the standard evaluator: its cell-type classifiers score 11.5% / 11.2% test accuracy.

MOG-DFM's guided demo runs at length 100 while reporting base FBD at 500 bp [3].

Does the innovation transfer?

DIRECTION

Yes

on every signature we can check.

The gates hold exactly, the ladder is monotone, expansion is reachable, the cost is unchanged, and the **null transfers too**: spread control does not raise band coverage in either state space, and in both it is the *contraction* rung that breaks the fidelity floor.

MAGNITUDE

2.6× smaller

a span of 0.20 against 0.52 in achieved spread. We attribute it to the property map, not the controller, and measured all three causes:

GC is affine in the simplex coordinates: its gradient is identical at every position with zero curvature, and the simplex renormalisation partly undoes the edit.

GC is a 200-position average, which suppresses the very variance the controller reads.

The band spans only 4.3 attainable GC values at granularity 1/200: a ceiling on what any method could move.

Ruled out: the trust region. It fired on 0–3% of steps, and the expanding cell clipped once in 25,600.

Takeaway. The evidence supports transfer of the *mechanism and its limits*, and locates the magnitude gap in a measurable property of the task. The lever for a larger effect is a **richer property**, such as a trained cell-type classifier logit used as classifier guidance rather than conditioning.

Limitations and failure modes

1 · The trust-region clip is a live confound at large $|w_{\text{eff}}|$

The plug-in rule clips on **884** sample steps where the contracting rung clips on **2,516**. At w_{eff} of order 10^3 the dispersion term sits far outside the trust region, so what reaches the sample is the clip's truncation of it. A rung can be *clip-limited* rather than *controller-limited*, and no metric here separates the two. **Fix:** a run at matched clip activity.

2 · The trade-off is re-parameterised, not removed

The chemistry cost tracks $|w_{\text{eff}}|$, not the *direction* of the knob, so strongly contracting and strongly expanding rungs both cost chemistry. This is a descriptive correlation over coupled cells, not a test.

3 · The batch is the estimator

V_b is computed over whatever tensor the sampler is handed, so running n samples in several batches runs *several independent controllers*. Our headline batch of 500 is a memory constraint, not a modelling choice, and we disclose it wherever n exceeds it.

4 · The flow has a dispersion drift of its own

The across-molecule variance *contracts and then re-expands* on **unguided** runs, so a setpoint on V_b is fighting a moving quantity.

Takeaway. Two of these are measured confounds, not speculation. They are the reason the controller claim is stated as provisional in 3e.

Conclusion and future directions

Back to the hypothesis. We asked whether making batch dispersion a *requested* quantity raises band coverage. At the power we have, **it does not**: no rung beats the plug-in rule at a corrected significance level. And we can say why, rather than only that.

The strongest supported conclusion

Servoing the *deviation* gain of a property-gradient guidance term against a batch-dispersion setpoint gives **monotone, two-sided, modality-portable** control of the generated batch's spread at **no additional sampling cost**. It converts the coverage-against-fidelity trade-off into a quantity the user *requests*, rather than removing it.

Mechanism and trade-off

The per-sample coefficient is affine in F_i , so BDG is the plug-in field at a rescaled weight aiming at a shifted target: it cannot leave that field's reachable set. It changes *which point* of that set the sampler lands on, and *how*. Holding the centring gain at w while servoing only the deviation gain buys contraction from the deviation mode, not by scaling the whole field, which is why BDG retains more chemistry at matched coverage.

Specific next experiments

Signed-w plug-in control, with our pre-registered drop rule: if it reaches BDG's spread at equal chemistry, we withdraw the controller claim and report the decomposition instead.

Matched clip activity, to separate clip-limited from controller-limited rungs.

A second seed pair for the open-loop replay.

A richer M2 property (cell-type classifier logit, nonlinear and motif-driven), to test whether the magnitude gap is the property map.

Bibliography

- [1] R. Ramakrishnan, P. Dral, M. Rupp, O. von Lilienfeld. Quantum chemistry structures and properties of 134 kilo molecules. *Scientific Data*, 2014. (QM9)
- [2] E. Hoogeboom, V. Satorras, C. Vignac, M. Welling. Equivariant Diffusion for Molecule Generation in 3D. *ICML*, 2022. (EDM; our stability and validity conventions)
- [3] T. Chen, Y. Zhang, S. Tang, P. Chatterjee. Multi-Objective-Guided Discrete Flow Matching for Controllable Biological Sequence Design. arXiv:2505.07086, 2025. (MOG-DFM; the enhancer data)
- [4] H. Stark, B. Jing, C. Wang, G. Corso, B. Berger, R. Barzilay, T. Jaakkola. Dirichlet Flow Matching with Applications to DNA Sequence Design. arXiv:2402.05841, 2024.
- [5] O. Davis, S. Kessler, M. Petrache, I. Ceylan, M. Bronstein, A. Bose. Fisher Flow Matching for Generative Modeling over Discrete Data. *NeurIPS*, 2024.
- [6] Y. Lipman, R. Chen, H. Ben-Hamu, M. Nickel, M. Le. Flow Matching for Generative Modeling. arXiv:2210.02747, 2022.
- [7] X. Liu, C. Gong, Q. Liu. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow. arXiv:2209.03003, 2022.
- [8] M. Albergo, N. Boffi, E. Vanden-Eijnden. Stochastic Interpolants: A Unifying Framework for Flows and Diffusions. *JMLR*, 2025.
- [9] V. Satorras, E. Hoogeboom, M. Welling. E(n) Equivariant Graph Neural Networks. *ICML*, 2021. (our backbone)
- [10] Y. Song, J. Sohl-Dickstein, D. Kingma, A. Kumar, S. Ermon, B. Poole. Score-Based Generative Modeling through Stochastic Differential Equations. *ICLR*, 2021.
- [11] T. Karras, M. Aittala, T. Aila, S. Laine. Elucidating the Design Space of Diffusion-Based Generative Models. *NeurIPS*, 2022.
- [12] H. Chung, J. Kim, M. McCann, M. Klasky, J. Ye. Diffusion Posterior Sampling for General Noisy Inverse Problems. *ICLR*, 2023. (DPS; our plug-in arm's ancestor)
- [13] Velocity-relative trust region: adopted prior practice, described on slide 2b; not claimed as ours.

Bibliography

- [14] J. Ho, T. Salimans. Classifier-Free Diffusion Guidance. arXiv:2207.12598, 2022.
- [15] H. Ye, H. Lin, J. Han, M. Xu, S. Liu, Y. Liang, J. Ma, J. Zou, S. Ermon. TFG: Unified Training-Free Guidance for Diffusion Models. *NeurIPS*, 2024.
- [16] H. Wang, Y. Liu, J. Chi, F. Liu, R. Xue, Y. Duan. CFG-Ctrl: Control-Based Classifier-Free Diffusion Guidance. arXiv:2603.03281, 2026.
- [17] F. Koulischer, F. Handke, J. Deleu, T. Demeester, L. Ambrogioni. Feedback Guidance of Diffusion Models. *NeurIPS*, 2025. arXiv:2506.06085.
- [18] E. Lempereur, N. Cuvelle-Magar, F. Coeurdoux, S. Mallat, E. Vanden-Eijnden. MGD: Moment Guided Diffusion for Maximum Entropy Generation. arXiv:2602.17211, 2026. (closest prior art)
- [19] I. Azangulov, L. Zhang, K. Ashouritaklimi. Variance-Tilted Diffusion Models for Diverse Sampling. arXiv:2606.22239, 2026.
- [20] G. Corso, Y. Xu, V. De Bortoli, R. Barzilay, T. Jaakkola. Particle Guidance: Non-I.I.D. Diverse Sampling with Diffusion Models. *ICLR*, 2024.
- [21] E. Ventura, B. Achilli, L. Ambrogioni, C. Lucibello. Negative guidance reduces means and expands variances in diffusion models. arXiv:2602.00716, 2026.
- [22] B. Boys, M. Girolami, J. Pidstrigach, S. Reich, A. Mosca, O. D. Akyildiz. Tweedie Moment Projected Diffusions for Inverse Problems. *TMLR*, 2024. (TMPD)
- [23] J. Song, Q. Zhang, H. Yin, M. Mardani, M.-Y. Liu, J. Kautz, Y. Chen, A. Vahdat. Loss-Guided Diffusion Models for Plug-and-Play Controllable Generation. *ICML*, 2023. (LGD-MC)
- [24] H. Ben-Hamu, O. Puny, I. Gat, B. Karrer, U. Singer, Y. Lipman. D-Flow: Differentiating through Flows for Controlled Generation. *ICML*, 2024.
- [25] S. Tang, Y. Zhang, A. Tong, P. Chatterjee. Gumbel-Softmax Flow Matching with Straight-Through Guidance. arXiv:2503.17361, 2025.
- [26] Y. Song, J. Gong, M. Xu, Z. Cao, Y. Lan, S. Ermon, H. Zhou, W.-Y. Ma. Equivariant Flow Matching with Hybrid Probability Transport for 3D Molecule Generation. *NeurIPS*, 2023. (EquiFM; appendix)
- [27] F. Bao, M. Zhao, Z. Hao, P. Li, C. Li, J. Zhu. Equivariant Energy-Guided SDE for Inverse Molecular Design. *ICLR*, 2023. (EEGSDE; appendix)

APPENDIX · NOT PART OF THE 15-MINUTE TALK

Q&A backup

Four topics the teaching team may ask about, one or two slides each.

A1

EquiFM, the borrowed generator

Architecture, probability path, and how our guidance arms run on it

A2

Property predictors

Our f_A and f_B against TFG's guide and oracle: architecture, training, accuracy

A3

The data split

Who trains on which half, and why halves

A4

The guidance arms

How each arm computes its correction, what it changes, and what it costs

EquiFM: the borrowed flow-matching generator

Released checkpoint	What it is
Paper	Song et al., NeurIPS 2023 [26]
Backbone	EDM-style E(n)-equivariant GNN dynamics [2, 9], width 256×9 layers
Parameters	5,339,920 (EMA weights)
State per atom	3 coordinates (Å) + 5 one-hot types ÷ 4 + 1 nuclear charge ÷ 10; explicit H, H/C/N/O/F
Training data	the full 100K QM9 training partition, unconditional

Hybrid probability path

Native time τ : 1 = noise, 0 = data.

Coordinates: an almost-linear path. The noise z is rotated (Kabsch) and permuted (Hungarian) to align with the molecule before interpolating, an equivariant optimal-transport coupling; the loss adds an angle penalty.

Atom types and charge: a variance-preserving path, β linear from 0.1 to 20.

Sampling: Euler from $\tau = 1$ to 0.

$$x_\tau = (1 - \tau) x_0 + s_x(\tau) z, \quad s_x = \epsilon + (1 - \epsilon)\tau$$

$$h_\tau = a h_0 + \sqrt{1 - a^2} z', \quad a = e^{-T(\tau)/2}$$

Our local pilot, unguided

256 molecules, 100 Euler steps: atom stability **98.37%**, molecule stability **86.72%**, validity **94.53%**, uniqueness 100%. A usability check, not a benchmark.

Why its chemistry beats our FM

Not isolated by us. It has about 2× our training data, 5.34M vs 3.75M parameters, one-hot scaled by $\frac{1}{4}$ (our unscaled one-hot is the leading suspect in 2a), and the aligned coordinate path.

Running our guidance arms on EquiFM

Why it is in the study. A second, external flow-matching generator asks whether the guidance comparison survives on a model we did not train. It is the v3 backend **equifm**, running all seven arms [26].

Endpoint estimate m , used by every arm	$m_x = (1 - \epsilon) x_\tau - s_x q_x, \quad m_h = a h_\tau + q_h/a$	Exact: algebra from its own regression targets; q_x, q_h are the raw heads.
Posterior covariance Σ , for TMPD and LGD-MC	$\Sigma = KJ, \quad k_x = \frac{s_x^2}{1 - \tau}, \quad k_h = \frac{1 - a^2}{a}$	Exact for atom types; approximate for coordinates , whose aligned noise is not an independent Gaussian channel.
Guided velocity	$\Delta v_x = -\frac{s_x}{1 - \tau} G_x, \quad \Delta v_h \approx -\frac{\beta(\tau)}{2} G_h$	The model's own velocity map, evaluated at the guided mean $m + K \cdot G$.

Property pair on this backend

TFG's released guide (tf_predict, time input 0) and oracle (evaluate) [15]. Its band is narrower than ours: $\delta = 0.157$ vs 0.175 (μ), 0.165 vs 0.506 (α), 0.0037 vs 0.0075 (gap). **In-band is never compared across backends.**

Disclosed limits

Trained on the full 100K, so it saw both halves. Only TMPD and LGD-MC use the Σ surrogate; plug-in, TFG and BDG do not. The charge channel is held fixed during guidance. 1.85× our memory per molecule.

Property predictors: our pair and TFG's

Our pair scores the fm and EDMsecond backends; TFG's pair scores EquiFM. f_A's **gradient** steers sampling; f_B only **scores** finished molecules.

	Ours: f_A and f_B	TFG guide (tf_predict)	TFG oracle (evaluate)
Network	invariant EGNN on squared distances, mean-pool, linear head [9]	EDM dynamics EGNN with an energy head [2, 15]	EDM property-classifier EGNN [2]
Width × layers	128 × 4	192 × 7	128 × 7
Edge attention	no: equal-weight sum over neighbours	yes	yes
Parameters (μ)	429,825 each	1,671,566	743,944
Inputs	coordinates + unscaled one-hot	coordinates + one-hot ÷ 8; time input fixed at 0	coordinates + one-hot + nuclear-charge powers
Training data	f_A: train_a · f_B: train_b (51,527 each)	QM9 second half (~50K)	QM9 first half (~50K)
Recipe	clean molecules; MSE on a standardised target; Adam 3e-4, cosine, batch 128, 120 epochs (~23 min); best validation MAE	clean molecules; EMA weights at epoch 2,000	best-validation checkpoint

Why TFG's are more accurate, and why we keep ours

MAE	μ (D)	α (Bohr ³)	gap (Ha)
our f_A	0.0872	0.2325	0.00402
our f_B	0.0877	0.2531	0.00374
TFG guide	0.0417	0.0925	0.00185
TFG oracle	0.0784	0.0824	0.00187

Same 3,000 calibration molecules. Ours are scored only on the half each did not train on; TFG's on the whole pool, which overlaps their training halves and flatters them.

The honest gap on μ . On molecules neither TFG net trained on: guide 0.0659 D, oracle 0.0728 D, against our f_B's 0.0840 D. About 20% better, not 2×.

Why theirs are more accurate

Capacity: 2–4× the parameters, 7 layers instead of 4.

Edge attention: a learned weight per neighbour, $a_{ij} = \sigma(\text{MLP}(m_{ij}))$; ours sums every neighbour equally.

Training length: the guide is the EMA at epoch 2,000 [15]; ours stop at 120.

Inputs: the oracle also reads nuclear-charge features.

Why we keep ours on our FM

Ours is disjoint **by construction**. TFG's is disjoint only by inference: its saved arguments are overwritten by its training script, and its oracle's exact training IDs are unknown. The cost is a wider band (ours / TFG δ : 1.12× μ , 3.07× α , 2.00× gap), which is why in-band never crosses backends.

Disjointness of TFG's pair, tested (μ , 3,000 molecules): guide wins by more than 2× on 30.9%, oracle on 40.9%, neither on 28.2%, against 37.4 / 37.4 / 25.3% predicted for disjoint halves. Replicated on α and gap.

The data split, and why it has this shape

Split	Trains or supplies	n
train_a	generator (FM and VP) and guide f_A	51,527
train_b	evaluator f_B only	51,527
val	checkpoint selection; molecule sizes	17,748
test	never used for training or selection	13,083

δ calibration: 3,000 molecules from train_a + train_b; each of our nets is scored only on the half it did not train on.

QM9 [1], 133,885 molecules, pinned by SHA-256; the split is fixed by seed 20260917.

1 · f_A and f_B never share data

Guidance is an optimiser aimed at f_A. Pushed hard, it finds inputs that fool f_A. The checker needs different blind spots, so the guide–evaluator gap stays a usable reward-hacking signal.

2 · The generator may share with f_A

The generator judges nothing. This keeps the claim to one sentence: f_B never saw a molecule the generator or the guide trained on.

3 · Halves, as the literature does

EDM's conditional protocol and TFG's EDMsecond train the generator on one ~50K half; TFG also shipped a full-data EDMfull and chose EDMsecond [2, 15]. Using both halves would set twice the data beside every number we compare to. A 70/30 split was rejected for the same reason.

The cost we accept: half the data. A published half-data EDM loses only ~0.3 atom-stability points against full data [2, 27], so the split does not explain our stability gap (2a).

The guidance arms, side by side

Shared frame: $m = x_t + (1 - t)v\theta \rightarrow$ frozen $f_A \rightarrow$ pull back through $J = \partial m / \partial x_t \rightarrow$ clip $\rightarrow$ add to the velocity ($t \geq 0.5, w = 1$).

Arm	Correction	What it changes	Passes / cell (2e)
DPS-style plug-in [12]	$G = \frac{y - f_A(m)}{s^2} J^\top \nabla f_A(m)$	the property at the endpoint; the baseline	10,000 / 4,000
TMPD-inspired [22]	$G = \frac{y - f_A(m)}{s^2 + g^\top \Sigma g} J^\top g, \quad \Sigma = k J, \quad g = \nabla f_A(m)$	shrinks the step where the endpoint is uncertain	10,000 / 4,000
LGD-MC [23]	$G = J^\top \sum_k \omega_k \frac{y - f_A(m + rz_k)}{s^2} \nabla f_A(m + rz_k), \quad r^2 = \frac{1}{d} \text{tr } \Sigma$	averages the likelihood over K draws around m	12,000 / 20,000
TFG [15]	$L(x) = \log \frac{1}{K} \sum_k \exp \left[- \left(\frac{f_A(x + \epsilon_k) - y}{\text{MAD}} \right)^2 \right]$ $\Delta_t = \rho \nabla_{x_t} L(m), \quad \Delta_0 : m \leftarrow m + \mu \nabla_m L(m) \quad (4 \text{ steps, no } J)$	adds steps on the predicted molecule itself	8,000 / 20,000
BDG (ours)	$\text{num}_i = (y - \bar{F}) - w_{\text{eff}} (F_i - \bar{F}), \quad w_{\text{eff}} = 1 + \eta e$	plug-in with a servoed deviation gain	10,000 / 4,000
D-Flow [24]	$\min_{x_0} \left(f_A(\Phi_\theta(x_0)) - y \right)^2, \quad \nabla_{x_0} = D\Phi_\theta(x_0)^\top \nabla f_A$	optimises the source noise through full ODE rollouts, each with a backward pass	[TODO]